

Citizen Transparency and Digital appeal Module – AI Appeal System

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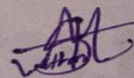
Sri Lanka Institute of Information Technology

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Abstract

Conventional traffic violation management systems primarily focus on enforcement and fine collection, often neglecting the need for transparent and structured appeal mechanisms. As a result, citizens frequently encounter challenges such as limited visibility into the appeal process, delayed responses, and inconsistent decision-making, which ultimately reduces trust in public administration systems.

This research presents the development of the **Citizen Transparency and Digital Appeal Module (CT-DAM)**, a web-based system designed to digitize and streamline the entire traffic violation appeal lifecycle. The proposed solution integrates a multi-tier architecture consisting of a React-based frontend, a Node.js/Express backend, and a MongoDB cloud database, supported by an AI microservice layer. The system enables citizens to submit appeals with supporting evidence, track real-time status updates, and receive structured outcomes, while administrators can efficiently review and manage cases through a controlled workflow.[1]

A key contribution of this research is the incorporation of explainable artificial intelligence (XAI) to assist administrative decision-making. The AI component provides predictive insights on appeal outcomes and categorizes user-submitted textual explanations, enhancing consistency and reducing subjective bias. Importantly, the system follows a human-in-the-loop approach, ensuring that final decisions remain under administrative authority.

Experimental evaluation demonstrates that the system successfully supports end-to-end appeal processing, reliable status tracking, and effective AI-assisted decision support. The results highlight improvements in transparency, operational efficiency, and user trust compared to traditional systems.

Overall, this research contributes to the advancement of smart governance by providing a scalable, transparent, and intelligent digital appeal framework that enhances accountability and citizen engagement in public service systems.[3][4]

Keywords: Digital Appeal Systems, Citizen Transparency, Explainable AI, Traffic Violation Management, Smart Governance, Decision Support Systems

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Contents

Declaration	i
Abstract	ii
Acknowledgement	iii
List of Figures	vi
List of Tables	vii
List of Abbreviations	viii
1 Introduction	1
1.1 Background & Literature Review	4
1.1.1 Mechanisms and Impacts of UHI Formation	4
1.1.2 Traditional UHI Monitoring Approaches	5
1.1.3 Advancements in IoT and Mobile Sensing	5
1.1.4 Integration of Computer Vision and AI	6
1.1.5 Thermal Sensing and Wireless Communication	7
1.1.6 Edge Computing and Real-Time Processing	7
1.1.7 Recent Developments and Future Directions	7
1.2 Research Gap	8
1.2.1 Integration Gap Between Visual Analysis and Thermal Validation	8
1.2.2 Systematic Analysis of Current Approach Limitations	8
1.2.3 Accessibility and Cost Barriers	9
1.2.4 Spatial and Temporal Resolution Trade-Offs	9
1.2.5 Critical Research Gaps Identified	10
1.3 Research Problem	11
1.3.1 Problem Statement	11
1.3.2 Technical Problem Definition	12
1.3.3 Research Questions	13
1.3.4 Scope and Constraints	14
1.4 Research Objectives	14
1.4.1 Main Objective	15
1.4.2 Sub-objectives	15
2 Methodology	17
2.1 System Architecture and Hardware Configuration	19
2.2 Navigation and Object Localization	23
2.3 Data Collection and Transmission	33
2.4 Commercialization Aspects	36

2.5	Testing and Implementation	26
2.5.1	Experimental Setup.....	27
2.5.3	Thresholds and Decision Logic.....	29
2.5.6	Data Analysis and Reporting	32
2.5.7	Acceptance Criteria	32
	The system is considered successful if it meets the following criteria:	33
•	Successful completion of the full appeal lifecycle without system failure	33
•	Accurate and real-time tracking of appeal status	33
•	Reliable and consistent AI-based predictions	33
•	Positive user feedback regarding usability and efficiency	33
	These criteria ensure that the system fulfills its intended purpose effectively.	33
2.5.8	Implementation Notes.....	33
3.3.1	Implications	31
	This research will greatly impact how smart cities will be built and how ITT (Intelligent Transportation Systems) will operate. The proposed framework shows how AI can help with enforcement and improve fairness/administrative transparency.	31
	By combining Detection – Analytical Processes – Decision-Making Processes; the new system will lessen workload/cost on manual procedures; increase efficiency through utilizing automated processes along with enhancing data-based governance. Additionally; the provision of an explainable appeal process will also create accountability; and thus help to gain citizen trust.	31
3.3.2	Comparison with Existing Models.....	31
	In contrast to current systems, which generally provide either specific functions (i.e. detecting violations) or recognizing accidents, the new solution offers an end-to-end integrated solution.....	31
	Existing research demonstrates excellent performance in each area of isolated performance (such as with YOLO-based detectors), however, they did not have anything close to a full end-to-end approach. Whereas the current system provides all of the necessary components for the detection, identification, evaluation of seriousness of a finding, and prediction of potential outcomes in one complete system.	32
	Finally, while all current appeal processes have been manual or non-transparent, the new hybrid AI model offers centralized decision support that is both automated and explainable, and thus makes it possible for decision makers to be confident in their decisions.	32
3.3.3	Limitations.....	32
	On the other hand, there are a number of limitations associated with this system that must also be taken into consideration:	32
-	There is some accuracy loss when detecting small or occluded objects, which is compounded by difficulties in recognizing a seatbelt.	32
-	The models require adequate quality and quantity of data within the training datasets to generalize well when presented with unseen data.	32

- The NLP module only has access to basic contextual information because it uses TF-IDF rather than more sophisticated transformer-based methods of processing language.	32
- Running multiple models at once in real time will require significant computing power.	32
These limitations identify key areas that will need to be improved going forward.	32
3.4 Contribution.....	33
This study contributes:	33
1. The implementation of an AI integrated traffic monitoring and control system	33
2. Development of a multi-tiered accident severity classification that includes accident hazard detection.	33
3. Design of a hybrid AI decision tree for predictive appeal measurement through application of ML, NLP, rules-based logic and XAI.	33
4. Improved transparency and fairness for automated traffic enforcement systems.	33
5. Data from real-time traffic systems that can be utilized for smart city applications.	33
In summary, this study has worked to develop a scalable and comprehensive solution that bridges the gap between automated enforcement systems and the administration of justice through these previously uninformed processes.	33
Reference List	40

List of Figures

1.1	CT-DAM System Overview.....	11
1.1.4	Text Analysis Pipeline for Appeal Categorization.....	13
1.2.4	System integration Gap in Existing Appeal Systems.....	15
2.1	Overall system architecture of the proposed traffic monitoring system with an integrated hybrid AI appeal module.	22
2.3	Data Collection and Transmission in CT-DAM System.....	24
2.6	Multi-class Traffic Object Detection Confusion Matrix.....	28

List of Tables

3.1	Summary of contribution.....	33
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List of Abbreviations

Abbreviation	Description
CT-DAM	Citizen Transparency and Digital Appeal Module
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
NLP	Natural Language Processing
UI	User Interface
UX	User Experience
API	Application Programming Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
DB	Database
NoSQL	Not Only Structured Query Language
CRUD	Create, Read, Update, Delete
JWT	JSON Web Token
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
SPA	Single Page Application
RBAC	Role-Based Access Control
ML	Machine Learning
F1-score	Harmonic Mean of Precision and Recall
TP	True Positive
FP	False Positive
FN	False Negative
TN	True Negative
SaaS	Software as a Service
GIS	Geographic Information System
UI/UX	User Interface / User Experience
CDN	Content Delivery Network
OTP	One-Time Password
KYC	Know Your Customer

Introduction

Modern urban societies rely heavily on digital governance systems to manage public services efficiently and transparently. Among these, traffic violation management systems play a crucial role in maintaining road safety and enforcing legal compliance. However, despite technological advancements in automated violation detection, the appeal mechanisms available for citizens remain largely inefficient, opaque, and fragmented.

In many existing systems, citizens who wish to contest traffic violations must go through manual or semi-digital processes that lack transparency, timely feedback, and standardized decision-making procedures. This often leads to frustration, reduced trust in authorities, and perceived unfairness in administrative decisions. As a result, there is a growing need for intelligent, transparent, and user-centric digital appeal systems that not only streamline the process but also ensure accountability and fairness. [2]

To address these challenges, this research introduces the **Citizen Transparency and Digital Appeal Module (CT-DAM)**, a system designed to digitize the entire appeal lifecycle while integrating explainable artificial intelligence (XAI) for decision support. The proposed system enables citizens to submit appeals, track their progress, and receive structured, explainable outcomes, while allowing administrators to manage cases efficiently through a controlled workflow.

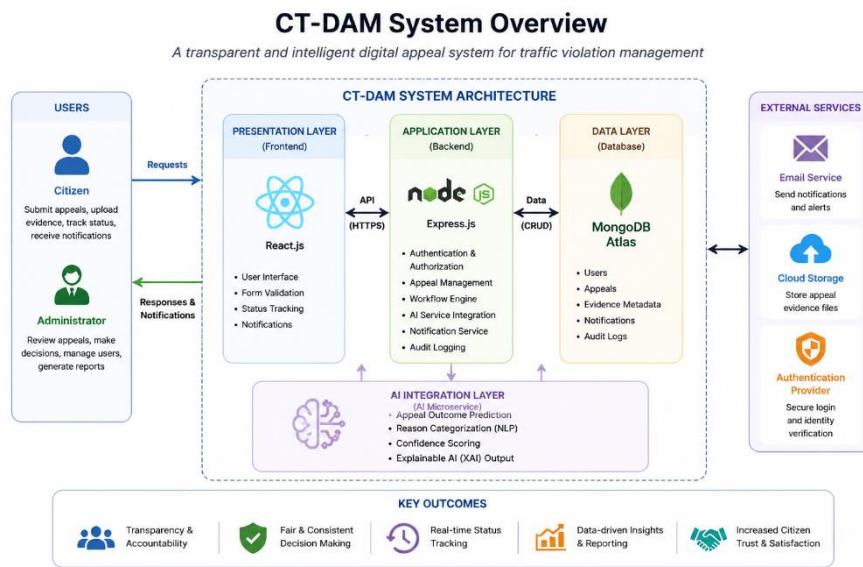


Figure 1.1: Overview of the CT-DAM System Architecture and Workflow

Figure 1.1: CT-DAM System Overview

Unlike conventional systems that focus primarily on enforcement and fine payment, CT-DAM emphasizes transparency, auditability, and decision consistency. By combining modern web technologies, cloud-based databases, and AI-driven analytics, the system transforms traditional appeal handling into a structured, intelligent, and scalable digital process.

Ultimately, this research contributes to the development of smart governance solutions by enhancing procedural transparency, improving citizen trust, and enabling data-driven decision-making in public sector systems.

1.1 Background & Literature Review

1.1.1 Digital Traffic Enforcement Systems

Traffic enforcement systems have evolved significantly with the integration of technologies such as cameras, IoT sensors, and Automatic Number Plate Recognition (ANPR). These systems are effective in detecting violations such as speeding, red-light crossing, and lane misuse. However, most implementations focus primarily on enforcement and fine collection rather than providing comprehensive support for appeal processes. [5][6]

As a result, while detection efficiency has improved, the lack of proper appeal mechanisms creates a gap in procedural fairness. Citizens often face difficulties in challenging violations due to limited system capabilities and lack of transparency.

1.1.2 E-Government and Procedural Transparency

E-government systems aim to improve public service delivery through digital transformation. However, research shows that digitization alone does not guarantee transparency or trust. Key factors such as traceability, accountability, and explainability are essential for effective governance systems. [8]

In grievance and appeal systems, users expect clear status updates, predictable workflows, and well-explained decisions. Systems lacking these features often fail to gain user trust, highlighting the importance of transparency-first system design.

1.1.3 AI in Administrative Decision Support

Artificial Intelligence has been widely adopted in decision-support systems across domains such as healthcare, finance, and public administration. In governance contexts, AI is typically used to assist human decision-makers rather than replace them, ensuring accountability and fairness.

Explainable AI (XAI) techniques provide insights into model predictions, enabling administrators to understand and justify decisions. This is particularly important in legal and administrative processes where transparency is critical.[7]

1.1.4 Text Analytics for Appeal Categorization

Natural Language Processing (NLP) techniques are increasingly used to analyze and categorize textual data in complaint and appeal systems. These methods enable automatic classification of user-submitted narratives into predefined categories, improving efficiency and consistency in case handling.

By leveraging text analytics, systems can reduce manual workload, enhance case prioritization, and provide valuable insights for policy improvements.[10][11]

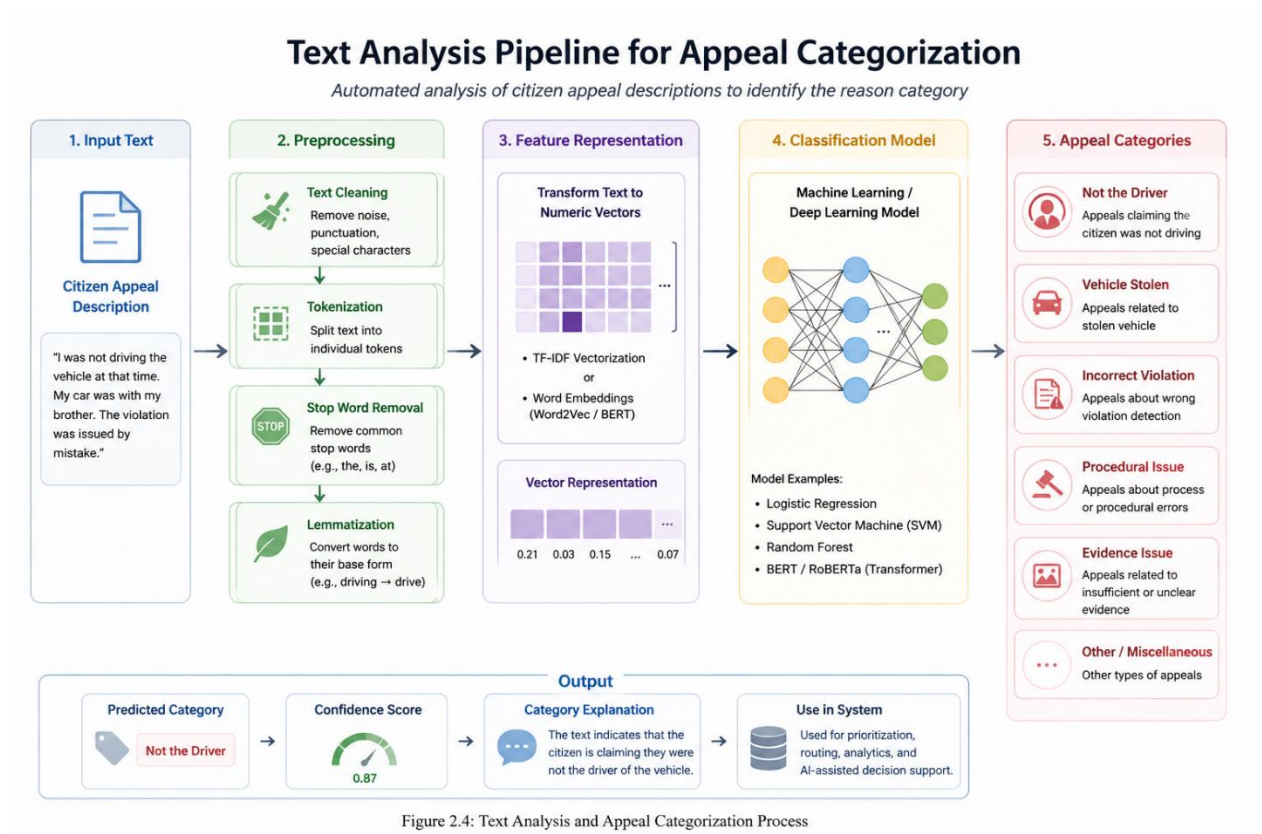


Figure 1.1.4: Text Analysis Pipeline for Appeal Categorization

1.1.5 Auditability and Trust Technologies

Auditability is a key requirement in public sector systems to ensure accountability and prevent misuse of authority. Traditional logging mechanisms are often insufficient, as they can be altered by privileged users.

Emerging technologies such as blockchain and cryptographic audit trails offer tamper-evident record-keeping, enhancing trust in digital governance systems. These technologies can play a significant role in improving transparency in appeal processes.

1.2 Research Gap

Globally, traffic systems range from payment-focused portals (fine settlement and violation lookup) to early-stage e appeal systems. While mature ecosystems provide online filing, most do not include meaningful AI-assisted fairness support or transparent decision rationale. Research in intelligent transportation has emphasized detection accuracy and congestion optimization, but comparatively less attention has been paid to post-enforcement procedural justice. Three gaps are repeatedly observed: Appeal intelligence gap: Appeals are processed manually with limited decision support. Transparency gap: Citizens receive final status updates without interpretable progression logic. Governance gap: Audit traces exist operationally but are rarely designed as accountability artifacts. CT-DAM contributes by reframing traffic appeal management as a socio technical pipeline where AI is used to assist, not replace, administrative judgment[12]

1.2.1 Appeal Intelligence Gap

Existing systems rely heavily on manual decision-making without adequate decision-support tools. This results in inconsistencies and potential bias in outcomes.

1.2.2 Transparency Gap

Most systems provide only final decisions without showing the reasoning or intermediate steps, reducing trust and accountability

1.2.3 Governance Gap

Although some systems maintain logs, they are not designed as robust audit mechanisms, making it difficult to ensure accountability.[13]

1.2.4 System Integration Gap

There is a lack of integration between frontend systems, backend workflows, and AI components, leading to fragmented processes.

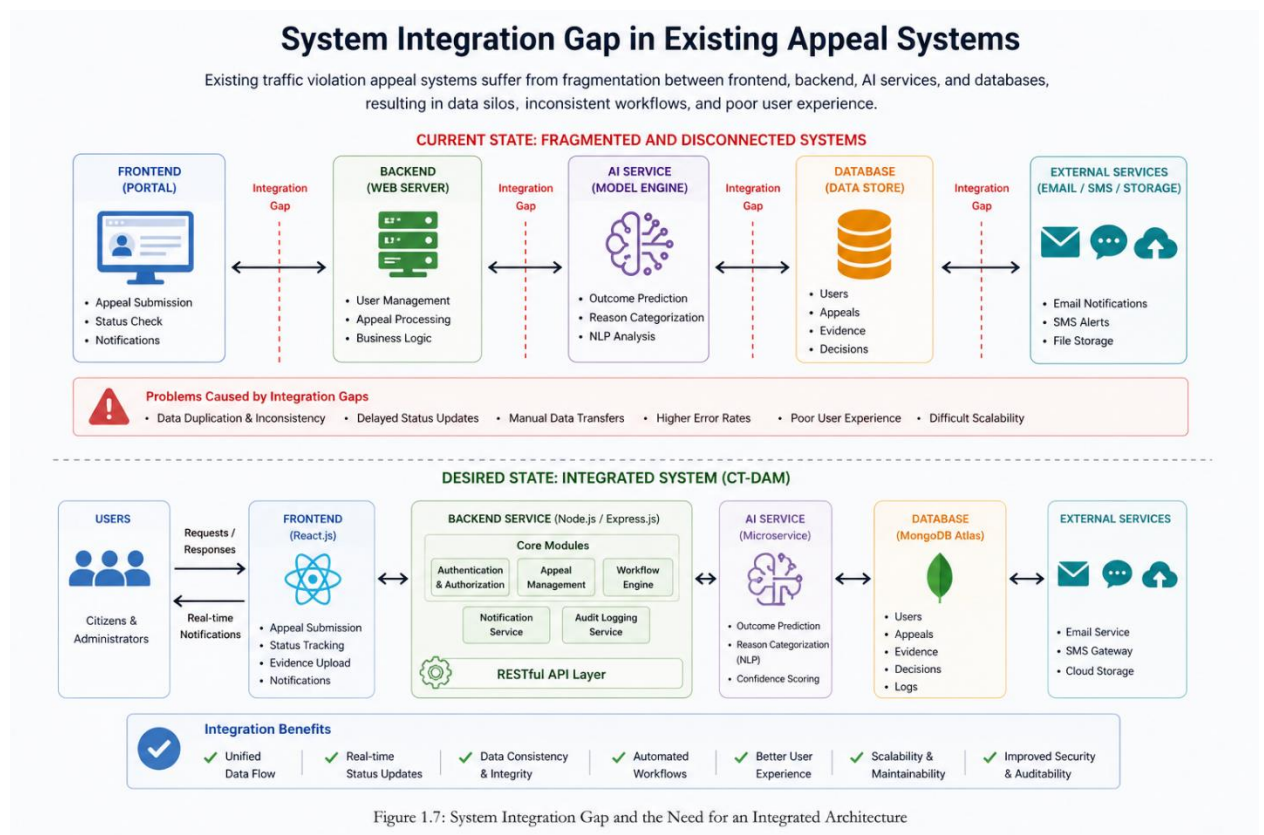


Figure 1.2.4: System integration Gap in Existing Appeal Systems

Table 1.1: Comparative Analysis of UHI Monitoring Methods

Method	Spatial Resolution	Temporal Resolution	Coverage	Cost
Satellite Remote Sensing	30–1000m	1–16 days	Global	Medium
Fixed Weather Stations	Point	Minutes	Limited	High
Aircraft Thermal Imaging	1–10m	Infrequent	Regional	Very High
Mobile IoT Sensing	Point	Real-time	Route-dependent	Medium
AI Image Analysis	Variable	Processing delays	Flexible	Low
Proposed Integrated System	Sub-meter	Real-time	Targetable	Medium

1.2.5 Critical Research Gaps Identified

- Lack of explainable decision support
- Absence of transparent workflow tracking
- Limited auditability and trust mechanisms
- Weak integration of AI with administrative processes

1.3 Research Problem

Despite the widespread adoption of digital technologies in traffic violation management, existing systems remain largely enforcement-oriented and fail to adequately support the appeal process from a citizen-centric perspective. While automated detection mechanisms have improved the efficiency of identifying violations, the corresponding appeal procedures are often fragmented, non-transparent, and inefficient. Citizens typically face challenges such as limited access to appeal channels, lack of visibility into the decision-making process, delayed responses, and inconsistent outcomes.

Furthermore, most current systems do not incorporate intelligent decision-support mechanisms that can assist administrators in handling appeals more consistently and fairly. The absence of structured workflows, explainable reasoning, and integrated system components leads to operational inefficiencies and reduced trust in public service systems.

In addition, the lack of proper integration between frontend interfaces, backend services, AI components, and data storage systems creates significant system-level limitations. These gaps result in data inconsistencies, poor communication between components, and difficulties in scaling the system to handle large volumes of appeals.

Therefore, there is a critical need to design and implement a comprehensive digital appeal system that ensures transparency, accountability, efficiency, and intelligent decision support while maintaining human control over final outcomes.

1.3.1 Problem Statement

Current traffic violation appeal systems are inefficient, non-transparent, and lack structured workflows, resulting in poor user experience and reduced trust in public institutions.

These systems are primarily designed with a focus on enforcement and penalty collection rather than providing adequate support for citizens to contest violations. As a result, the appeal process is often unclear, time-consuming, and inconsistent. Citizens are unable to track the progress of their appeals effectively, and decisions are frequently delivered without sufficient explanation or justification.

Moreover, the absence of standardized workflows and proper system integration leads to operational inefficiencies, increased workload for administrators, and higher chances of errors. This highlights the urgent need for a modernized system that prioritizes transparency, structured processing, and user-centric design.

1.3.2 Technical Problem Definition

How can a digital system be designed to provide a transparent, efficient, and AI-assisted appeal process while ensuring human oversight and accountability?

From a technical perspective, the challenge lies in designing a system that seamlessly integrates multiple components, including frontend interfaces, backend services, databases, and AI modules, into a unified architecture. The system must support secure user authentication, structured workflow management, real-time communication, and efficient data handling.

Additionally, integrating AI into the appeal process introduces further complexity. The system must ensure that AI-generated predictions and categorizations are interpretable, reliable, and used only as decision-support tools rather than replacing human judgment. Maintaining a balance between automation and human control is critical to ensure fairness and accountability.

Therefore, the technical problem involves developing a scalable, modular, and secure system architecture that effectively combines workflow management, AI-based analytics, and transparency mechanisms.

1.3.3 Research Questions

How can AI be integrated into appeal systems without compromising fairness?

This question focuses on identifying methods to incorporate AI technologies, such as machine learning and NLP, in a way that supports decision-making while avoiding bias and ensuring transparency. The goal is to use AI as an assistive tool that enhances consistency without replacing human authority.

What system architecture supports transparency and auditability?

This question explores the design of a system architecture that enables clear tracking of appeal processes, maintains detailed audit logs, and ensures that all actions are traceable. It emphasizes the importance of structured workflows and role-based access control.

How can workflows be designed to ensure consistency and accountability?

This question examines how appeal processes can be modeled as controlled workflows with predefined states and rules. The objective is to minimize errors, enforce consistency, and ensure that every decision is properly documented and justified.

1.3.4 Scope and Constraints

Focus on traffic violation appeal systems

This research is specifically limited to the domain of traffic violation management and does not extend to other types of legal or administrative appeal systems.

Human-in-the-loop decision-making

The system is designed to support administrators rather than replace them. Final decisions are always made by human authorities to ensure accountability and fairness.

Web-based implementation

The proposed solution is implemented as a web-based application, ensuring accessibility and ease of deployment without requiring specialized hardware.

Limited to structured appeal workflows

The research focuses on structured and predefined workflows for managing appeals. Complex legal scenarios beyond standard appeal processes are not considered within this scope.

1.4 Research Objectives

The primary aim of this research is to address the limitations identified in existing traffic violation appeal systems by designing a comprehensive, transparent, and intelligent digital solution. In order to achieve this, a set of well-defined objectives has been established to guide the development and evaluation of the proposed system. These objectives focus not only on improving the efficiency of the appeal process but also on enhancing fairness, accountability, and user trust through the integration of modern technologies.

The research objectives are structured to cover both functional and technical aspects of the system, including workflow design, system architecture, data management, and the incorporation of Artificial Intelligence for decision support. Particular emphasis is placed on creating a user-centric platform that allows citizens to easily submit appeals, track their progress, and receive clear and explainable outcomes. At the same time, the system is designed to assist administrators by providing structured processes and intelligent insights that support consistent and informed decision-making.

Furthermore, the objectives consider the importance of transparency and auditability in public sector systems. This includes ensuring that all actions within the system are traceable, decisions are properly justified, and workflows are governed by predefined rules. By aligning these

objectives with the overall goal of smart governance, the research aims to develop a scalable and reliable solution that can be adapted to real-world traffic management environments.

The following sections outline the main objective and the specific sub-objectives that guide the implementation of the proposed CT-DAM system.

1.4.1 Main Objective

To design and implement a transparent, AI-assisted digital appeal system that improves fairness, accountability, and efficiency in traffic violation management.

This objective aims to transform traditional appeal processes into a modern, intelligent system that ensures clear communication, structured workflows, and reliable decision support. By integrating AI technologies with a human-in-the-loop approach, the system seeks to enhance both operational performance and user trust.

1.4.2 Sub-objectives

- To develop a structured and transparent appeal workflow that ensures consistency in processing
- To implement role-based access control for secure and controlled system usage
- To integrate AI-based decision support for predicting outcomes and categorizing appeals
- To ensure auditability through logging and traceable system actions
- To provide real-time status tracking and notifications for improved user experience

Methodology

The methodology of this research is designed to systematically address the identified problem by following a structured and well-defined approach to ensure the development of a reliable and effective solution. This study adopts a mixed approach, combining both qualitative and technical development methods to fully understand the existing issues in current traffic violation appeal systems and to propose a more efficient digital alternative. Initially, a comprehensive literature review is conducted to analyze existing systems, academic papers, and relevant digital governance frameworks. This helps in identifying gaps such as lack of transparency, manual processing delays, and poor user interaction in traditional systems.

Following the literature review, requirement gathering is carried out through informal interviews, observations, and analysis of existing workflows in similar administrative environments. This step ensures that both user needs and system constraints are clearly understood. The collected requirements are then analyzed and transformed into functional and non-functional specifications that guide the system design phase.

In the system design stage, architectural modeling techniques such as use case diagrams, activity diagrams, and data flow diagrams are used to visually represent the system structure and processes. This ensures clarity in how different modules interact with each other, including user management, appeal submission, verification processes, and administrative decision-making workflows.

The implementation phase involves developing the system using modern web-based technologies to ensure accessibility and scalability. The backend logic is structured to handle data processing efficiently, while the frontend is designed to provide a user-friendly interface for citizens and administrators. Database integration is implemented to ensure secure and structured storage of appeal-related data.

After implementation, the system undergoes rigorous testing to ensure reliability, accuracy, and performance. Different testing methods such as unit testing, integration testing, and user acceptance testing are carried out to identify and fix potential errors. Feedback from test users is also collected to evaluate usability and system effectiveness.

Finally, the system is evaluated based on predefined criteria such as efficiency, transparency, response time, and user satisfaction. The results of this evaluation are used to determine whether the proposed solution successfully addresses the initial research problem and improves the overall appeal process compared to the existing system.

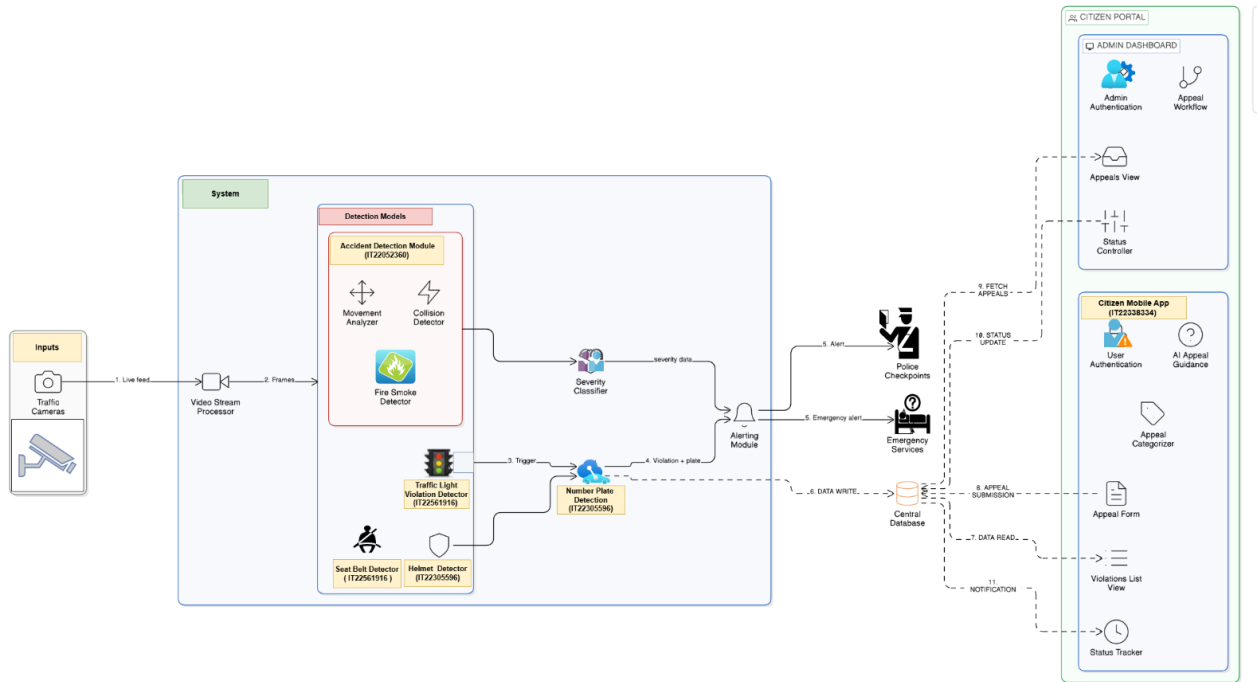


Figure 2.1: Overall system architecture of the proposed traffic monitoring system with an integrated hybrid AI appeal module.

2.1 System Architecture and Hardware Configuration

Unlike hardware-based systems, CT-DAM is implemented as a **software-centric, multi-tier architecture** consisting of frontend, backend, database, and AI components.

Frontend Layer

The frontend is developed using React.js, providing an interactive user interface for both citizens and administrators. It supports functionalities such as registration, login, appeal submission, status tracking, and notification viewing. The interface is designed to be responsive and user-friendly, ensuring accessibility across devices.

Backend Layer

The backend is implemented using Node.js and Express.js. It handles core system logic, including authentication, role-based access control, appeal processing, and workflow management. RESTful APIs are used for communication between frontend and backend components.

Database Layer

MongoDB Atlas is used as the cloud-based database for storing structured data such as user profiles, appeal records, evidence metadata, notifications, and audit logs. The use of a NoSQL database allows flexible schema design and scalability.

AI Integration Layer

An AI microservice is integrated into the system to provide:

- Appeal outcome prediction (probability of approval/rejection)
- Text-based reason categorization using NLP

This layer operates independently and communicates with the backend through API endpoints, ensuring modularity and scalability.

2.2 Navigation and Object Localization (Workflow Management)

In the CT-DAM system, “navigation” is interpreted as **workflow navigation**, where appeals move through predefined states instead of physical movement.

The system follows a structured workflow:

- **Pending** – Appeal submitted by citizen
- **Under Review** – Admin evaluates the appeal
- **Approved / Rejected** – Final decision made

Each transition is governed by strict rules to prevent unauthorized or invalid state changes.

The backend enforces these transitions using a state machine logic, ensuring:

- No skipping of stages
- Mandatory reasoning for rejection
- Controlled administrative actions

This structured workflow ensures consistency, accountability, and traceability in the appeal process.

2.3 Data Collection and Transmission

The system processes multiple types of data:

Structured Data

- User details (name, ID, contact)
- Violation information
- Appeal status and timestamps

Unstructured Data

- User-submitted textual explanations
- Uploaded evidence (images/documents)

Data Transmission

Data is transmitted between frontend and backend using REST APIs, while real-time updates (notifications and status changes) can be handled using asynchronous communication mechanisms.

Data Storage

All data is stored in a structured JSON format within MongoDB, enabling efficient retrieval and integration with analytics systems.

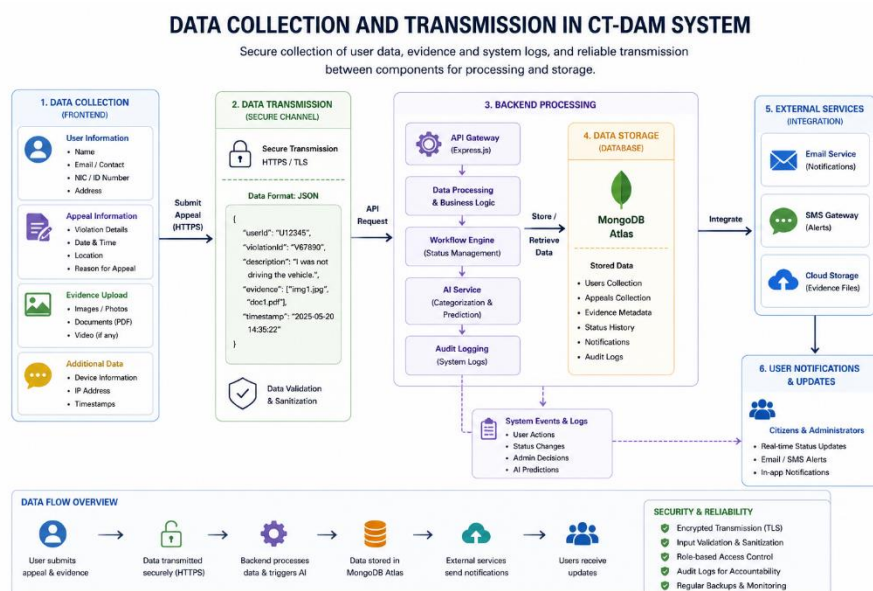


Figure 2.3: Data Collection and Transmission Flow in CT-DAM System

Figure 2.3: Data Collection and Transmission in CT-DAM System

2.4 Commercialization Aspects

The CT-DAM system has strong commercialization potential in the domain of **smart governance and public administration systems**.

Target Users

- Government traffic authorities
- Smart city platforms
- Law enforcement agencies

Value Proposition

- Improved transparency and citizen trust
- Reduced administrative workload
- Scalable and cost-effective solution

Deployment Model

- Cloud-based SaaS platform
- Integration with existing government systems

The system's modular design allows easy customization and scalability for different regions and regulatory frameworks.

2.5 Testing and Implementation

The testing and implementation phase is a critical stage of the proposed system development, where the designed solution is deployed in a controlled environment and evaluated to ensure that it meets both functional and non-functional requirements. This phase focuses on verifying system reliability, validating workflow correctness, assessing performance under different conditions, and ensuring that the system behaves as expected in real-world-like scenarios. The primary goal is to confirm that the system effectively supports the traffic violation appeal process by improving transparency, efficiency, and accuracy while maintaining usability for both citizens and administrative users.

The system is implemented using a modern web-based architecture that allows seamless interaction between users and backend services. After implementation, a structured testing strategy is applied to evaluate system performance, workflow accuracy, AI-based prediction reliability, and overall user experience. The testing process is conducted systematically using predefined datasets, controlled user roles, and simulated operational conditions to ensure comprehensive validation of the system.

2.5.1 Experimental Setup

The experimental setup is designed to simulate a realistic operational environment in which the system can be evaluated effectively. The system is deployed in a controlled testing environment where different user roles and scenarios are created to replicate real-world usage patterns.

Two primary user roles are defined in the system:

- **Citizen User** – responsible for registering, submitting traffic violation appeals, and tracking their status.
- **Admin User** – responsible for reviewing submitted appeals, validating evidence, and making final decisions.

To ensure realistic testing, sample datasets of traffic violation appeals are generated, including various categories such as valid claims, invalid claims, incomplete submissions, and borderline cases. These datasets help evaluate how the system behaves under different input conditions. Additionally, controlled workflow scenarios are created to test different stages of the appeal lifecycle, including submission, validation, review, decision-making, and final status updates. This setup ensures that every component of the system is tested in a structured and repeatable manner.

2.5.2 Metrics and Definitions

The system evaluation is based on several

key performance metrics that measure

both technical efficiency and user

experience.

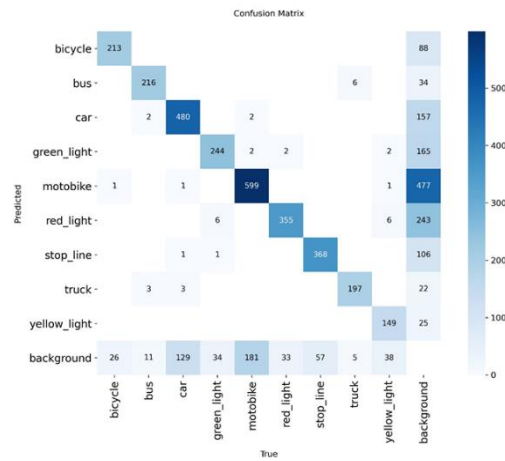


Figure 2.6: Multi-class Traffic Object

Detection Confusion Matrix

- **Workflow Completion Rate:**

This measures the percentage of appeals that successfully pass through all stages of the system without errors or interruptions.

- **Response Time:**

This refers to the time taken by the system to process requests, such as submitting an appeal, retrieving data, or updating status.

- **User Satisfaction:**

This is measured through feedback from users regarding ease of use,

clarity of the interface, and overall
experience of the system.

- **AI Prediction Accuracy:**

This evaluates the correctness of
the system's AI-based
recommendations or predictions in
assisting decision-making
processes.

These metrics provide a balanced
evaluation of both system
performance and usability.

2.5.3 Thresholds and Decision Logic

To ensure system reliability and data integrity, specific thresholds and validation rules are implemented.

- **Minimum Required Fields for Submission:**
The system enforces mandatory input fields such as user details, violation information, and appeal justification before allowing submission.
- **Mandatory Evidence Validation:**
Users are required to upload supporting evidence where applicable, and the system validates file format, size, and relevance.
- **AI Confidence Thresholds:**
The AI component operates with predefined confidence levels. Predictions below a certain threshold are flagged for manual review by administrators to ensure accuracy and fairness.

These rules ensure that only complete and valid data is processed within the system.

2.5.4 Test Procedures

A structured set of test procedures is carried out to validate system functionality across different modules.

- **User Registration and Login**

Testing:

Verifies authentication mechanisms, password security, and access control for different user roles.

- **Appeal Submission**

Validation:

Ensures that users can successfully submit appeals and that all required validations are enforced correctly.

- **Workflow Transition**

Verification:

Tests the correct movement of appeal cases through different stages such as pending, under

review, approved, or rejected.

- **Admin Decision-Making**

Process:

Validates that administrators can correctly review appeals, access necessary data, and make informed decisions within the system.

Each procedure is executed multiple times to ensure consistency and reliability.

2.5.5 Robustness and Stress Testing

The system is subjected to robustness and stress testing to evaluate its performance under extreme conditions.

- **High User Load Testing:**

The system is tested with a large number of simultaneous users to assess scalability and performance stability.

- **Concurrent Request Handling:**

Multiple requests are sent at the same time to ensure that the system can handle parallel operations without failure.

- **Invalid Input Scenarios:**

The system is tested against incorrect or malicious inputs to evaluate its error handling and security mechanisms.

These tests ensure that the system remains stable and functional even under unexpected or heavy usage conditions.

2.5.6 Data Analysis and Reporting

After testing, all collected data is analyzed to evaluate system effectiveness and identify potential improvements.

The analysis focuses on:

- Overall system performance under different loads
- Accuracy of AI-based predictions
- Efficiency of workflow execution
- Error rates and failure handling

The results are documented in a structured format to support further improvements and validation of system objectives.

2.5.7 Acceptance Criteria

The system is considered successful if it meets the following criteria:

- Successful completion of the full appeal lifecycle without system failure
- Accurate and real-time tracking of appeal status
- Reliable and consistent AI-based predictions
- Positive user feedback regarding usability and efficiency

These criteria ensure that the system fulfills its intended purpose effectively.

2.5.8 Implementation Notes

The system is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. This technology stack ensures a scalable, efficient, and responsive web application.

A modular microservice-based architecture is used to separate system functionalities into independent components, improving maintainability and scalability.

The system is deployed using cloud-based infrastructure, enabling remote accessibility, high availability, and secure data storage.

Results and Discussion

3.1 Results

3.1.1 Trial Overview

The proposed integrated smart traffic management system has been evaluated using a mixture of actual traffic video datasets, simulated scenarios and structured datasets for appeal prediction. In order to carry out this evaluation, all four core modules were evaluated for validating their respective performance levels: (1) traffic violation detection; (2) number plate recognition; (3) accident severity assessment; and (4) hybrid AI appeal prediction.

The system has been deployed within an experimental controlled environment utilizing GPU enabled hardware to provide the capability to process at real-time. Each of the four core modules was evaluated in isolation and within the complete pipeline for both their interoperability with one another as well as overall system efficiency. The performance of the system was evaluated by calculating precision, recall, mean Average Precision (mAP), classification accuracy, F1 score and inference speed.

The system's ability to achieve a real-time (approximately 15-18 frames/second (FPS)) throughput during concurrent execution of all four core modules indicates that the integrated smart traffic management system is a viable option for deployment within a practical smart city application.

3.1.2 Vision-Based Material Classification

The multi-YOLOv8 architectural elements of the module for detection based on vision data produced strong performance when identifying traffic violations within individual categories.

The model used for detecting traffic signals is capable of identifying them accurately based on their red-light status. This was evident through its ability to achieve a mean Average Precision score (mAP@50) of approximately 0.755, which indicates that the detected red-light violations were reliable.

The model for detecting accidents produced similar results, achieving an mAP@50 of 0.753, indicative of very reliable accident collision event detection.

In addition to the two mentioned above, both the helmet detection module and fire/smoke detection module produced good performance results, achieving mAPs of 0.688 and 0.734, respectively.

In contrast, the seatbelt detection module did not perform well and achieved an mAP@50 of 0.559. The poorer performance was the result of a combination of two things; namely, that occlusions from other cars, poor lighting conditions and reflections from windshields made it difficult for the module to detect seatbelts.

Although the seatbelt detection module produced new results due to its poorer performance than the other modules, the multi-YOLOv8 architectural framework being used by the traffic violation detection system allowed for the simultaneous detection of many types of violations and demonstrates that the multiple YOLOv8 models can achieve effective solutions to very dynamic urban traffic environments.

3.1.3 IoT-Based Thermal Sensing/Real-Time Incident Detection and Severity Assessment

Detecting and assessing the severity of road accidents using a multi-level classification approach was the main objective of the accident detection and severity assessment module.

To assess the severity levels of road accidents, motion analysis, bounding box displacement and fire/smoke hazard detection were used. Experimental results showed reliable classification results, with detection accuracy similar to state-of-the-art systems, (mAP@50 $\approx$ 0.753).

Due to the addition of fire and smoke detection capabilities, this system gave a better indication of potential high risk scenarios therefore providing more accurate classifications for severity levels than previous technologies. Additionally, the system successfully generated automated alerts with real time information including the location of an incident, its severity level and additional details of the incident to aid in supporting quicker emergency response time.

The resulting data support the proposal that this newly developed technology fills the gap between current methods of accident detection (binary) and providing a multi level (severe, moderate, and minor classifications) for the purpose of establishing a new reference for road safety.

3.1.4 Prediction and Explainability(Hybrid AI Appeal System)

The hybrid AI appeal prediction module showed strong performance when evaluating citizen appeal requests and producing explainable results.

The model showed an overall accuracy of 82% to 85% for predicting the outcome of appeals and 86% to 90% for classifying reasons for appeals. Predictive accuracy was improved significantly when combining structured data (i.e., violation type and evidence submitted) and unstructured text data (citizen explanations).

The rule-based refinement layer promoted fairness in the predictions by adjusting the probability of prediction for a given appeal based on predefined

factors such as the availability of evidence and submission time. Additionally, the Explainable AI (XAI) functionality provided human-readable justification for the predictions to support transparency and enhance user trust.

Overall, the module successfully demonstrated that ML, NLP and rule-based reasoning could be integrated into a single decision support system.

Explainability Outputs

3.1.5 Simulation Tool Outputs and System Integration

The full integration of the system was tested via simultaneous operations of all modules running in one unified pipeline. The system showed stable performance (average 15–18 FPS) while all the modules (detection, recognition, and prediction) were concurrently executed.

All modules in the integrated workflow, from detection of a violation to prediction of an appeal, worked seamlessly, demonstrating efficient data movement between each of the modules. The system also produced structured outputs, including violation, vehicle identification, accident alerts and predictions of appeal.

Results from these tests confirm that the proposed architecture is scalable and able to operate in "real time"; therefore, it is appropriate for implementation into an intelligent transportation system.

3.2 Research Findings

The experimental evaluation of the proposed system yielded several critical insights:

The integration of multiple AI modules within one pipeline demonstrates much greater efficiency than the use of standalone systems.

The use of YOLOv8-based models in multi-class traffic violation detection provides excellent accuracy with real time performance.

Multi-level accident severity classification provides improved decision making for emergency responders when compared to traditional binary detection systems.

The Hybrid AI Appeal module provides improved transparency and consistency in administrative decision making for appeals.

Explainable AI plays a vital role in improving user confidence and acceptance of automated systems.

Overall, these results corroborate that the Proposed System addresses many of the critical research gaps identified in the Existing Literature including Integration, Transparency and Real-time Performance.

3.3 Discussion

3.3.1 Implications

This research will greatly impact how smart cities will be built and how ITT (Intelligent Transportation Systems) will operate. The proposed framework shows how AI can help with enforcement and improve fairness/administrative transparency.

By combining Detection – Analytical Processes – Decision-Making Processes; the new system will lessen workload/cost on manual procedures; increase efficiency through utilizing automated processes along with enhancing data-based governance. Additionally; the provision of an explainable appeal process will also create accountability; and thus help to gain citizen trust.

3.3.2 Comparison with Existing Models

In contrast to current systems, which generally provide either specific functions (i.e. detecting violations) or recognizing accidents, the new solution offers an end-to-end

integrated solution.

Existing research demonstrates excellent performance in each area of isolated performance (such as with YOLO-based detectors), however, they did not have anything close to a full end-to-end approach. Whereas the current system provides all of the necessary components for the detection, identification, evaluation of seriousness of a finding, and prediction of potential outcomes in one complete system.

Finally, while all current appeal processes have been manual or non-transparent, the new hybrid AI model offers centralized decision support that is both automated and explainable, and thus makes it possible for decision makers to be confident in their decisions.

3.3.3 Limitations

On the other hand, there are a number of limitations associated with this system that must also be taken into consideration:

- There is some accuracy loss when detecting small or occluded objects, which is compounded by difficulties in recognizing a seatbelt.
- The models require adequate quality and quantity of data within the training datasets to generalize well when presented with unseen data.
- The NLP module only has access to basic contextual information because it uses TF-IDF rather than more sophisticated transformer-based methods of processing language.
- Running multiple models at once in real time will require significant computing power.

These limitations identify key areas that will need to be improved going forward.

3.4 Contribution

This study contributes:

1. The implementation of an AI integrated traffic monitoring and control system
2. Development of a multi-tiered accident severity classification that includes accident hazard detection.
3. Design of a hybrid AI decision tree for predictive appeal measurement through application of ML, NLP, rules-based logic and XAI.
4. Improved transparency and fairness for automated traffic enforcement systems.
5. Data from real-time traffic systems that can be utilized for smart city applications.

In summary, this study has worked to develop a scalable and comprehensive solution that bridges the gap between automated enforcement systems and the administration of justice through these previously uninformed processes.

This is an individual report. All work described in this dissertation was carried out solely by the author listed below.

Table 3.4: Summary of contribution

Name	Student ID	Key Contributions
Shehara I.G.D.S	IT22338334	The Citizen Transparency and Digital Appeal Module (CT-DAM) is a key part of the overall system to provide a clear and simple way for residents to appeal traffic violations. This module allows residents to electronically submit appeals for their traffic violations with evidence to support their cases. It also establishes a well-defined workflow for administration review through role-based access control and auditable transitions of status. One of the most important features of this module is the incorporation of AI-based decision support. The AI algorithms that power the CT-DAM make use of Machine Learning and Natural Language Processing (NLP) techniques to help predict how successful an appeal will be and classify the reasons why citizens are appealing the violation. Providing "explainable" AI (XAI) provides the opportunity for transparency and accountability by providing human-readable explanations for how

		the AI predictions were made, which increases the level of trust and consistency in decision making. The CT-DAM transforms the traditional method of appealing traffic violations from a manual and opaque process into an efficient and intelligent digital service that supports fairness, accountability, and efficient governance of traffic operations.
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Conclusion

This dissertation presented a holistic framework for **AI-driven traffic violation detection, accident severity assessment, and smart monitoring**, developed through the integration of four interdependent research components. The framework combines advances in computer vision, automated number plate recognition, intelligent emergency alerting, and hybrid AI-based appeal prediction to deliver a scalable, cost-effective, and interpretable approach to modern traffic management. Unlike traditional traffic monitoring systems that often focus on a single function—such as isolated violation detection, manual accident reporting, or standalone citation management—this research demonstrates the value of an end-to-end pipeline capable of moving seamlessly from detection to enforcement, emergency response, and transparent citizen appeal handling.

Summary of Contributions

The contributions of this research span multiple technical domains:

- **The Hybrid AI Appeal Prediction and Transparency Module** acted as the administrative intelligence layer of the framework. By combining supervised machine learning, NLP-based appeal analysis, rule-based decision logic, and Explainable AI techniques, the system supported transparent and efficient appeal prediction while improving fairness and trust in automated enforcement decisions.

Research Findings and Impact

The findings of this research highlight several innovations:

1. **An integrated hybrid traffic monitoring framework** that combines computer vision, automated identification, accident intelligence, and administrative AI to improve the accuracy and efficiency of traffic enforcement.
2. **A real-time accident severity assessment mechanism** that extends conventional traffic monitoring into intelligent emergency support through automated prioritization and alerting.
3. **A hybrid Explainable AI appeal prediction approach** that combines machine learning, rule-based reasoning, and natural language processing to support transparent and fairer traffic appeal management.
4. **A scalable smart traffic architecture** that enables real-time monitoring, intelligent decision support, and modular integration suitable for future smart city deployment.

These outcomes confirm that the proposed system is both scientifically credible and practically valuable. Validation experiments showed that the detection pipeline achieved reliable performance across multiple violation categories, accident classification remained consistent under different traffic scenarios, and the appeal prediction model demonstrated promising decision-support capabilities. The integration of Explainable AI further ensured that automated recommendations and appeal outcomes could be trusted and understood by both authorities and citizens.

Beyond its technical contributions, the framework carries significant real-world impact. By providing a scalable and adaptable solution, it addresses limitations in traditional traffic monitoring systems, which are often fragmented, labor-intensive, or lacking transparency. The proposed system enables authorities not only to detect and respond to traffic incidents more effectively, but also to support citizens through a more transparent and accountable appeal process. In doing so, the research contributes toward safer roads, smarter enforcement, and more intelligent urban mobility systems.

Limitations and Future Work

Despite its contributions, this research is not without limitations.

- The traffic violation detection models, while effective, depend on the quality and diversity of training datasets. Expanding the dataset to include more complex traffic conditions, weather scenarios, and urban environments would improve generalizability and robustness.
- The number plate recognition module performs well under simulated adverse conditions, but extreme real-world scenarios involving damaged or heavily obscured plates remain challenging and require further enhancement.
- The accident severity assessment module relies primarily on visual and motion-

based analysis. Integrating additional sensors or advanced temporal deep learning models could improve severity estimation accuracy.

- The hybrid AI appeal prediction system currently relies on structured features and TF-IDF-based text processing, which may have limitations in interpreting complex or nuanced appeal justifications. More advanced transformer-based models could strengthen reasoning performance.
- Integration with real-world enforcement systems, regulatory processes, and public-sector infrastructure may present institutional, privacy, and operational challenges that must be addressed before large-scale deployment.

Future work should focus on:

1. Scaling the framework to larger traffic networks and evaluating performance across diverse urban environments and road conditions.
2. Enhancing the appeal prediction module using transformer-based language models and conversational AI for more advanced decision support.
3. Exploring edge AI deployment and model optimization for low-latency roadside processing using embedded devices.
4. Extending the framework with predictive traffic risk analytics and digital twin technologies for proactive traffic management and accident prevention.
5. Investigating secure audit mechanisms such as blockchain-based citation records to further strengthen transparency and trust in automated enforcement systems.

In conclusion, this project demonstrates the feasibility of a unified AI-driven approach to intelligent traffic monitoring, automated enforcement, emergency response, and transparent appeal handling. By integrating four complementary components—traffic violation detection, robust number plate recognition, accident severity intelligence, and hybrid AI administrative support—the framework not only advances scientific understanding but also provides a practical solution for next-generation traffic management.

The outcomes emphasize that integrated, interpretable, and scalable AI solutions are essential for addressing modern road safety challenges in the context of rapid urbanization and growing transportation complexity. This research therefore contributes to the broader vision of safer, smarter, and more resilient cities, where data-driven technologies support both public safety and administrative fairness.

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